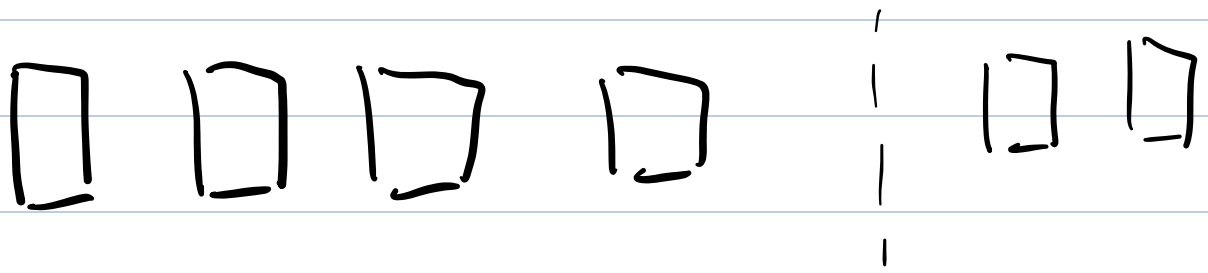
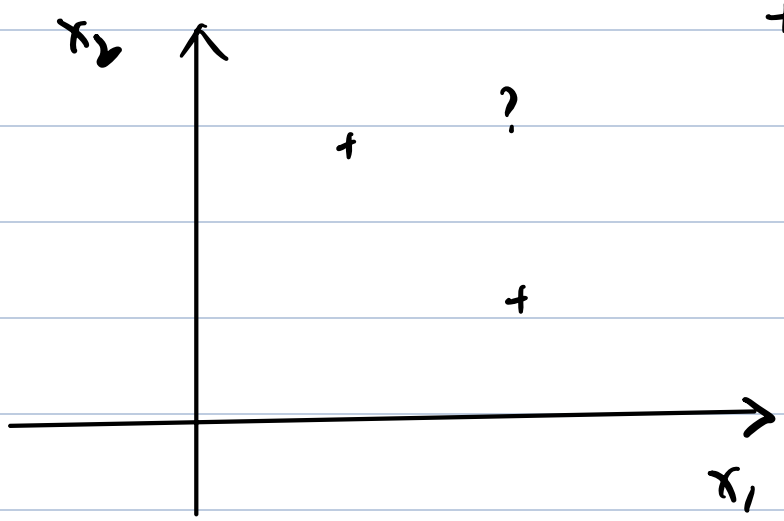
Data processing

feature vector : Stores the features of the data.



Training set : Test set

Linear classifier



(labels)
+ (Data)

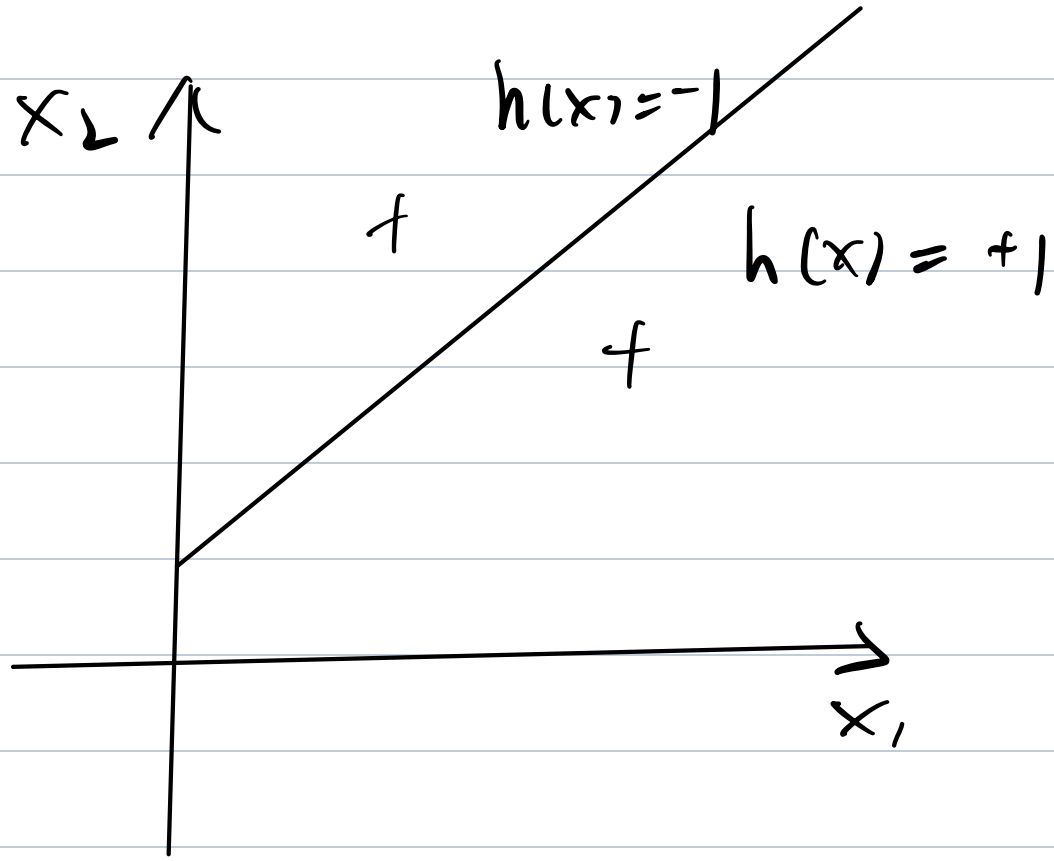
(unknown labels)
? : (Data)

S_n : training sets (A set of examples that includes the feature vector and the corresponding label.

Test Set: A set of examples that includes the feature vectors and unknown labels.

Classifier: mapping from points to labels.

$h : x \to \{-1, 1\}$



Training error $E_n(h) = \frac{1}{n} \sum [h(x^{(i)}) \neq y^{(i)}]$

$= 1$ if error

0 otherwise

Test error $E(h)$ .....

$E_2 = 0$

Generalization: A generalize from training set to testing set.

hypothesis space: The collection of all potential models that can be created using a particular learning algorithm to solve a specific problem.

Supervised learning Tasks.

① Muti-way classification

② Regression

③ Structured prediction

Types of machine learning

- Supervised learning
  - prediction based on examples of correct behavior
- Unsupervised learning
  - no explicit target, only data, goal to model/discover
- Semi-supervised learning
  - supplement limited annotations with unsupervised learning
- Active learning
  - learn to query the examples actually needed for learning
- Transfer learning
  - how to apply what you have learned from A to B
- Reinforcement learning
  - learning to act, not just predict; goal to optimize the consequences of actions
- Etc.

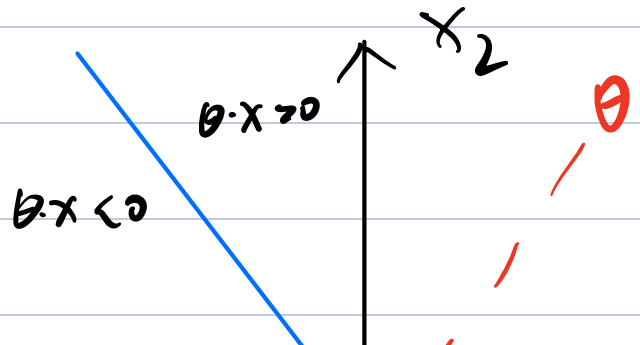
Linear classifier

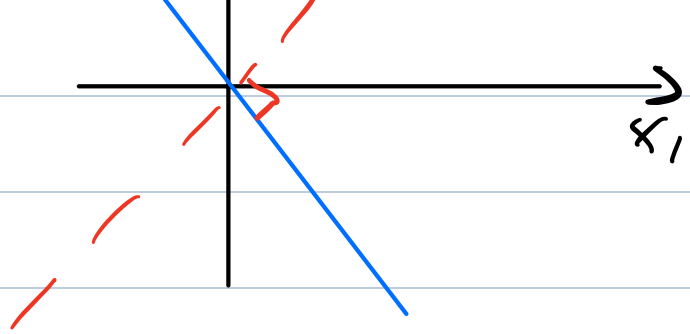
through origin: $\theta_1 x_1 + \theta_2 x_2 = 0$

$\theta = \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix}$, $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

$\Rightarrow \theta \cdot x = 0$

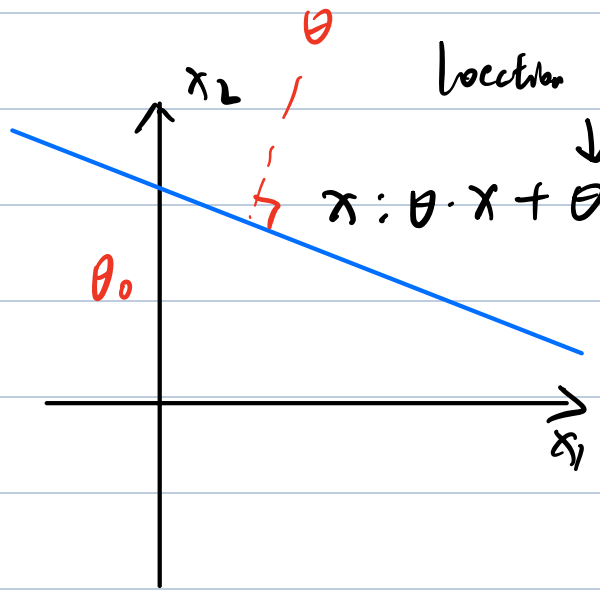
$\Rightarrow$ θ vector is orthogonal to all the points on the decision boundary.





$$h(x;\theta) = \text{sign}(\theta \cdot x)$$
$$\theta \in R^d$$

general:



location of boundary

$\downarrow$

$x : \theta \cdot x + \theta_0 = 0$

vector θ orthogonal to decision boundary

$$h(x;\theta;\theta_0) = \text{sign}(\theta \cdot x + \theta_0)$$

Linear Separation

Definition: Training examples $S_n = \{(x^{(i)}, y^{(i)}), i = 1 \ldots n\}$ are linearly separable if there exists a parameter vector $\hat{\theta}$ and $\hat{\theta}_0$ such that $y^{(i)} \cdot (\hat{\theta} \cdot x^{(i)} + \hat{\theta}_0) > 0$

for all $i = 1, \ldots, n$

Training Error

through origin: $E_n(\theta) = \frac{1}{n} \sum_{i=1}^{n} [y^{(i)} \cdot \theta \cdot x^{(i)} \le 0]$

if true $= 1$

otherwise $= 0$

General: $E_n(\theta; \theta_0) = \frac{1}{n} \sum_{i=1}^{n} [y^{(i)} \cdot (\theta x^{(i)} + \theta_0) \le 0]$

Perceptron Algorithm

① Start: $\theta = 0$ (vector), $\theta_0 = 0$ (scalar

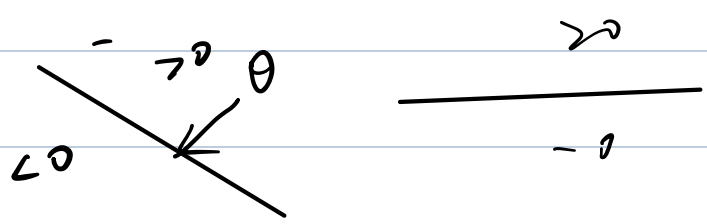
② For all $i = 1, \ldots, n$

if $y^{(i)} \cdot (\theta \cdot x^{(i)} + \theta_0) \le 0$ then

$\theta = \theta + y^{(i)} \cdot x^{(i)}$, $\theta_0 = \theta_0 + y^{(i)}$

∵ later updates may make the earlier examples no longer correct.

∴ go through multiple times.



Pseudo Code:

```
procedure Perceptron({(x^(i), y^(i)), i=1 ... n}, T)
    θ = 0 (vector)
    for t = 1 ... T do
        for i = 1 ... n do
            if y^(i) (θ · x^(i) + θ0) ≤ 0 then
                θ = θ + y^(i) · x^(i)
                θ0 = θ0 + y^(i)
    return θ, θ0
```

Average Perceptron Algorithm

1. $\theta = 0$, $\theta_0 = 0$, $\theta_{sum} = 0$, $c = 0$
2. 对于每个训练样本 (x_i, y_i)

   if $y^{(i)} \cdot (\theta \cdot x^{(i)} + \theta_0) < 0$

   $\theta \leftarrow \theta + x^{(i)} \cdot y^{(i)}$

   $\theta_{sum} \leftarrow \theta_{sum} + \theta$

   $c \leftarrow c + 1$
3. 计算平均权重

   $$W_{avg} = \frac{W_{sum}}{c}$$